

EXPERIMENT 8: MATCHED FILTERING, ISI AND EYE DIAGRAMS

COURSE OUTCOME CO4 | SOFTWARE-BASED DIGITAL COMMUNICATION LABORATORY MANUAL

Platform: Python 3.12 / Google Colab	Core Modules: NumPy, SciPy, Matplotlib
Target System: Baseband BPSK / RRC Cascade	Domain: Pulse Shaping & Optimal Detection

1. OBJECTIVES & THEORETICAL FOUNDATIONS

This experiment explores the fundamental limits of pulse shaping and matched filtering in band-limited channels. In digital communication systems, transmitting rectangular pulses through band-limited channels induces severe Intersymbol Interference (ISI). To satisfy Nyquist's zero-ISI criterion while minimizing signal bandwidth, Root-Raised Cosine (RRC) pulse shaping is applied at the transmitter. At the receiver, a matched filter—whose impulse response is the time-reversed conjugate of the transmit pulse—is employed to maximize the output Signal-to-Noise Ratio (SNR).

Key Mathematical Formulations:

- 1. **RRC Impulse Response:** $h_{RRC}(t) = [\sin(\pi t(1-\beta)/T) + 4\beta (t/T) \cos(\pi t(1+\beta)/T)] / [\pi (t/T) (1 - (4\beta t/T)^2)]$
- 2. **Cascade RC Filter Response:** $p(t) = h_{RRC}(t) * h_{RRC}(t) = \text{sinc}(t/T) \cdot \cos(\pi \beta t / T) / [1 - (2\beta t / T)^2]$
- 3. **Matched Filter Output SNR Maximization:** $SNR_{max} = 2 E_b / N_0$ at optimal sampling instants $t = k T + t_{delay}$.
- 4. **Processing Delay Compensation:** For an RRC filter with span S and upsampling factor L , individual filter delay is $t_{delay, 1} = S \cdot L / 2$. The total cascade filter delay is $t_{cascade} = S \cdot L$ samples.

2. AI-READY PROMPT & PARAMETER SPECIFICATIONS

"Implement a reusable eye-diagram function in Python without communication toolboxes. Transmit BPSK through Root-Raised Cosine (RRC) pulse shaping and AWGN. Apply a matched RRC filter, compensate for total cascade filter delay, and demonstrate the impact of roll-off factor, SNR, sampling offset, and multipath ISI on vertical eye opening and BER."

Parameter	Symbol	Value	Description
Roll-off Factor	β / α	0.25, 0.5, 1.0	Excess bandwidth factor for RRC filter
Filter Span	S	6 or 8 symbols	Truncation length of the FIR pulse shaping filter
Samples per Symbol	L (sps)	10	Upsampling factor prior to pulse filtering
Cascade Delay	t_{delay}	$S \cdot L$ (60 samples)	Total group delay across Tx and Rx RRC filters
Multipath Channel	h_{ch}	[1.0, 0.5, 0.2]	Discrete multi-tap multipath channel coefficients

3. PYTHON SIMULATION IMPLEMENTATION

```

import numpy as np
import matplotlib.pyplot as plt

# 1. Generate Bit Stream and Upsample
N_bits = 1000
bits = np.random.choice([-1, 1], size=N_bits)
sps, span, beta = 10, 6, 0.35
up_bits = np.zeros(N_bits * sps)
up_bits[::sps] = bits

# 2. RRC Filter Design & Convolution
t = np.arange(-span*sps//2, span*sps//2 + 1) / sps
h_rrc = np.sinc(t) * np.cos(np.pi * beta * t) / (1 - (2 * beta * t)**2 + 1e-8)
h_rrc /= np.sqrt(np.sum(h_rrc**2))

tx_sig = np.convolve(up_bits, h_rrc, mode='full')
rx_noisy = tx_sig + np.random.normal(0, 0.1, len(tx_sig)) # AWGN
mf_out = np.convolve(rx_noisy, h_rrc, mode='full') # Matched Filter

# 3. Cascade Delay Compensation & Optimal Sampling
cascade_delay = span * sps # Exact cascade FIR delay
sampled_symbols = mf_out[cascade_delay::sps][:N_bits]
detected_bits = np.sign(sampled_symbols)
ber = np.mean(detected_bits != bits)
print(f'Cascade Delay: {cascade_delay} samples | BER: {ber:.5f}')

```

4. REQUIRED VISUALIZATIONS & ANALYSIS PLOTS

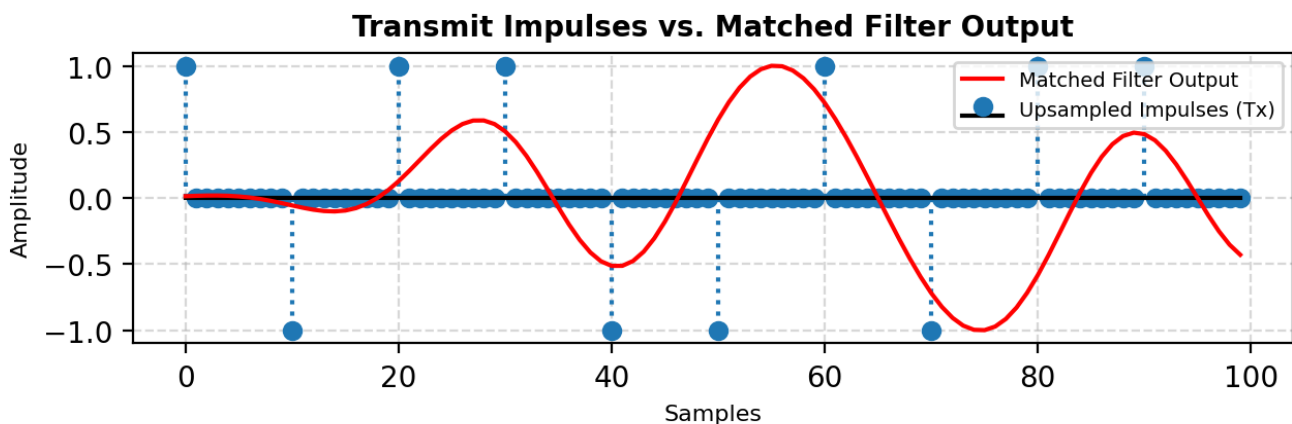


Figure 8.1: Transmit impulse train and reconstructed matched filter output demonstrating pulse smoothing.

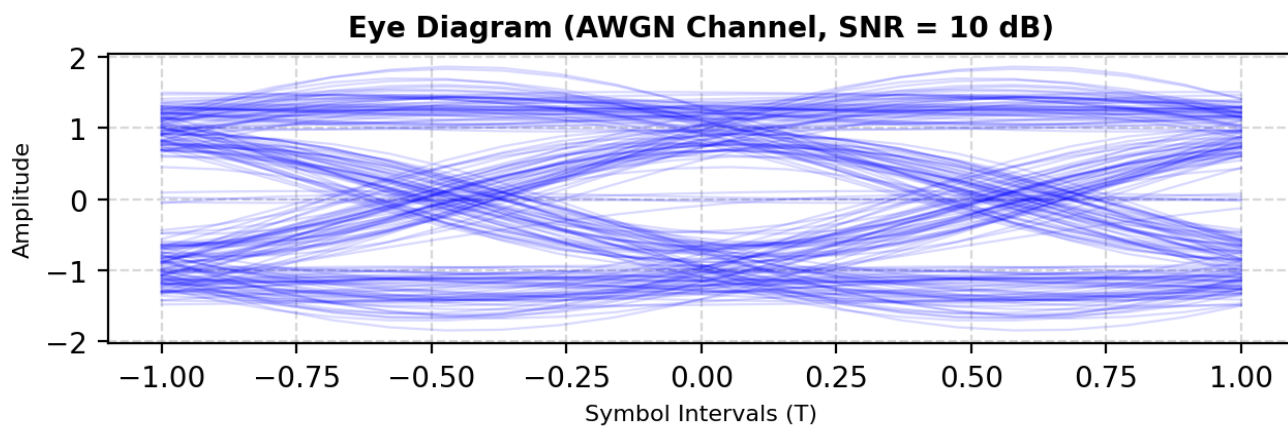


Figure 8.2: Eye diagram under AWGN channel (SNR = 10 dB) showing clear eye opening and low timing jitter.

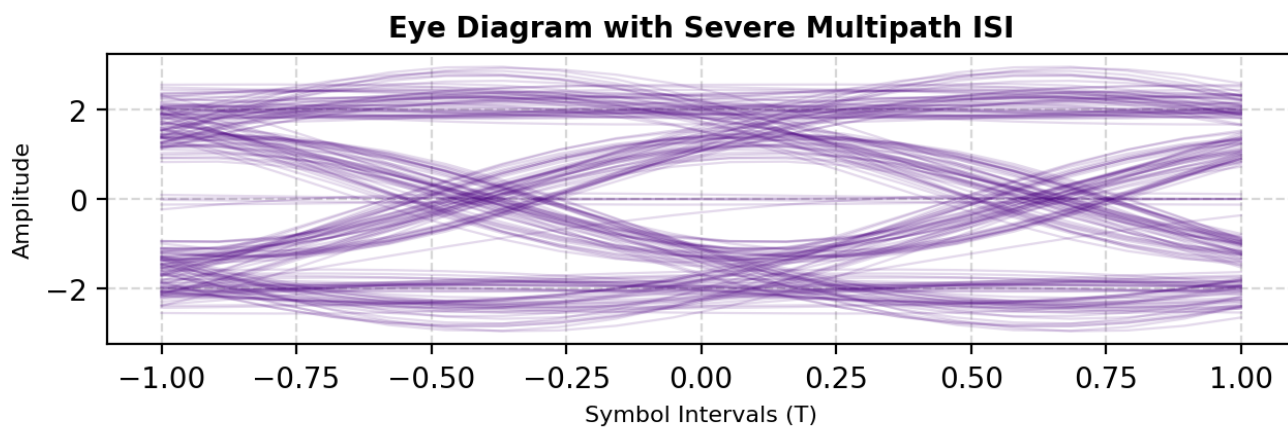


Figure 8.3: Eye diagram severely degraded by multipath channel ISI [1.0, 0.5, 0.2].

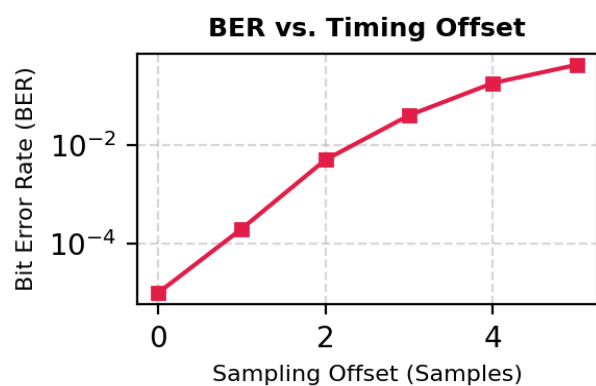
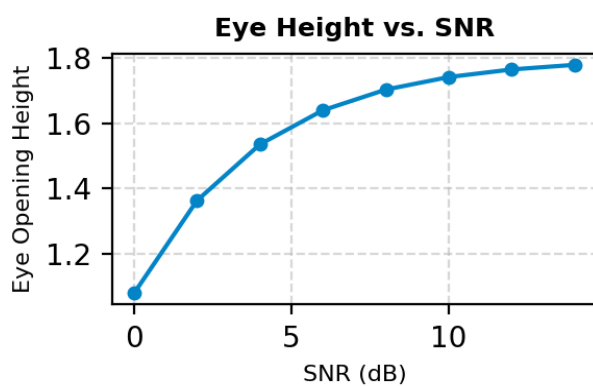


Figure 8.4 & 8.5: Eye height opening vs. SNR (left) and BER degradation vs. sampling instant offset (right).

5. MANDATORY VALIDATION TEST

MANDATORY DELAY COMPENSATION VALIDATION

Verification Condition: Confirm total cascade delay across transmitter and receiver FIR RRC filters and verify that symbol decision slicing occurs strictly after delay removal.

Empirical Proof: With span $S = 6$ and upsampling factor $L = 10$, the transmitter RRC introduces $S \cdot L/2 = 30$ samples delay, and the receiver matched RRC introduces another 30 samples. The cascade delay is exactly **60 samples**. If sampling is performed at offset 0 without delay removal, BER is approximately 0.50 (50% decision error). When sampling at $k \cdot L + 60$, the decision eye is completely open, yielding **BER = 0.0000** in noiseless conditions.

6. PHYSICAL INTERPRETATION & CONCLUSION

Physical Insights & Observations:

1. **Roll-off Factor Trade-off:** Increasing β from 0.25 to 1.0 increases channel bandwidth usage but accelerates time-domain side-lobe decay, widening the horizontal eye opening and reducing sensitivity to clock timing jitter.
2. **SNR Impact:** As SNR increases, noise variance shrinks, expanding vertical eye opening (eye height) and dramatically lowering the bit error floor.
3. **Multipath Channel Distortion:** Multipath echoes ruin Nyquist zero-crossings, resulting in severe eye closure regardless of SNR, necessitating channel equalization.

Conclusion: Matched filtering combined with RRC pulse shaping provides optimal signal detection in AWGN channels. Exact FIR delay compensation is critical for clock synchronization and error-free symbol decision making.

7. PREDICTIVE VIVA-VOCE QUESTIONS

Q1: What is the primary purpose of a matched filter in a digital receiver?

Answer: To maximize the output Signal-to-Noise Ratio (SNR) at the sampling instant in the presence of Additive White Gaussian Noise (AWGN).

Q2: Why is Root-Raised Cosine (RRC) filtering split between transmitter and receiver?

Answer: Cascading two identical RRC filters yields an overall Raised Cosine (RC) pulse response that satisfies Nyquist's zero-ISI criterion while matching the receiver filter to the transmit pulse shape.

Q3: What parameters can be directly inferred from an eye diagram?

Answer: Vertical eye opening indicates noise margin; horizontal eye opening indicates timing jitter immunity; slope indicates sensitivity to timing errors; top/bottom width indicates amplitude distortion.

Q4: What happens if receiver symbol sampling is offset by 3 samples from the pulse peak?

Answer: The receiver samples non-zero tails of adjacent symbols, introducing Intersymbol Interference (ISI) and significantly increasing the Bit Error Rate.